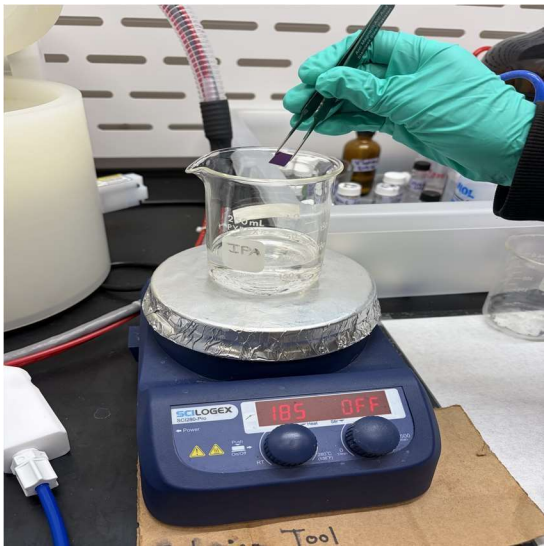
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UCSC Science Internship Program 2026 |

July - August 2026

Two-dimensional (2D) materials offer promising alternatives to silicon for smaller, faster, and energy-efficient electronic devices due to their unique electrical, optical, and mechanical properties. The internship focused on fabricating 2D material-based devices on silicon dioxide substrates. I gained hands-on experience in graphene mechanical exfoliation, chemical vapor deposition (CVD) growth, polymer transfer stamp fabrication, material transfer, and characterization—developing practical knowledge of key 2D device fabrication techniques.

1. Preparing the chip



I began by cleaning silicon-based surfaces with isopropyl alcohol to prepare them for the graphene transfer process.

2. Preparing the Tape



I used the Scotch Tape Method to separate thin graphene flakes from graphite by repeatedly folding and peeling the tape. I marked darker regions containing flakes for later use.

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3. Transfer to SiO2 Chips



I aligned the marked areas of the tape with cleaned silicon-based chips, pressed and heated them, and slowly peeled away the tape to transfer the graphene flakes onto the surfaces.

4. Optical Identification of Flakes



I examined the prepared samples under an optical microscope, checking the size and color of the graphene flakes. I systematically inspected each chip and recorded the locations of suitable flakes for future experiments.

5. Preparing PPC Chips



I prepared silicon-based surfaces by coating them with a thin layer of plastic polymer. Using a spin coater, I spread the coating evenly across each surface before heating it to prepare it for the next stage.

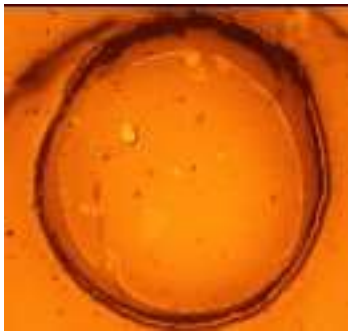
6. Preparing the stamp



I fabricated the transfer stamp by cutting glass slides into small pieces and placing two circular silicone discs of different sizes on top of each other. I secured the discs to the glass using adhesive tape, taking care not to press directly on the raised surface.

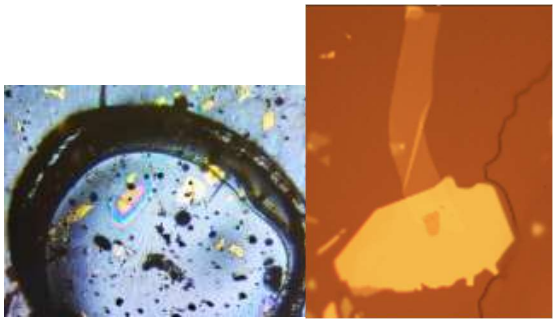
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7. PPC to stamp

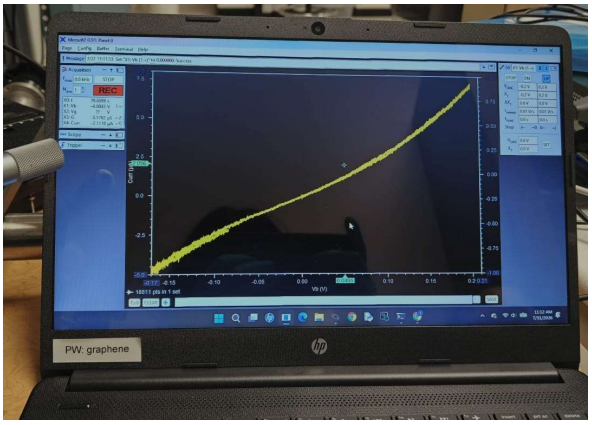


I trimmed the edges of the polymer-coated substrate and used a piece of Scotch tape with a circular opening to carefully lift the thin polymer film from the surface. I then transferred the film onto the prepared silicone stamp for use in the material-transfer process.

8. Dry transfer process



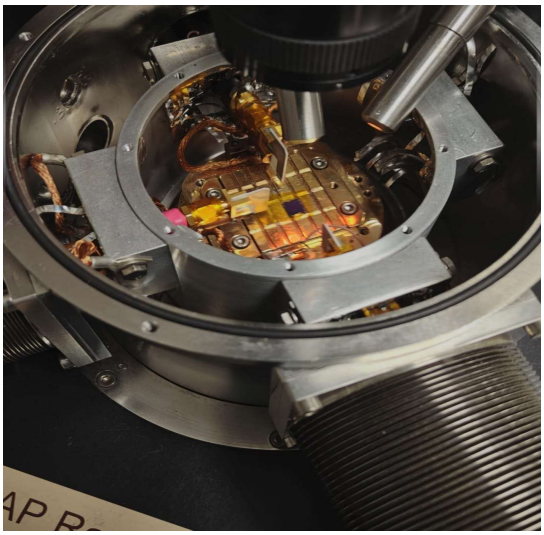
I used the transfer stamp to pick up and stack three ultra-thin materials: hexagonal boron nitride, graphene and a transition metal dichalcogenide. I identified contact between the stamp and the material through the appearance of rainbow-colored patterns. Once contact was established, I introduced a small amount of water and slowly raised the stage, allowing the polymer film to lift the selected flake from its surface.

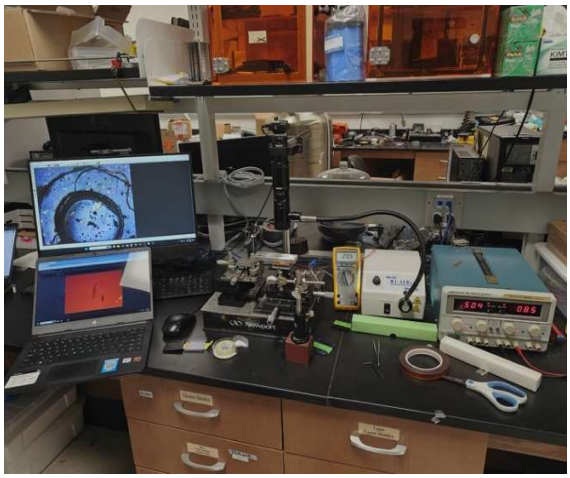




By changing the state of the device, we plotted the voltage vs resistance graph of each chip to see whether it has the suitable electrical and structural properties.

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This internship gave me valuable insight into how scientific research is conducted in a professional laboratory environment. Alongside the practical work, reading recent scientific literature strengthened my understanding of the principles behind the experiments. Connecting these papers to the work in the laboratory helped me understand not only how each technique was performed, but also why it was necessary and how it influenced the final device.

Overall, the internship strengthened my technical skills and confidence in a research environment while deepening my appreciation for the precision and attention to detail required in experimental science. It also reinforced my interest in pursuing further opportunities in materials science, nanotechnology, and experimental physics.